Course: Selected Topics in Networks (ECE5306) - 7th Assignment
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1 Overview: Two 2960 switches (SW1, SW2), one 2911 router (R1) and six PCs serve three departments: CS (VLAN 10), EE (VLAN 20) and BA (VLAN 30). Each department has one PC on each switch, so every VLAN spans both switches.

Key technologies:

- **VLANs:** Each department gets its own VLAN, creating separate broadcast domains and logical isolation without extra hardware. Inter-department traffic can only pass through the router, where it can be controlled.
- **802.1Q trunking:** Because the VLANs exist on both switches, the inter-switch links (Gi0/1, Gi0/2) and the SW1 Fa0/24 to R1 Gi0/0 link are trunks. Each frame is tagged with its VLAN ID, so all three VLANs (10, 20, 30) share the same links.
- **Router-on-a-stick:** Different VLANs use different subnets, so a Layer 3 device is needed. R1's single Gi0/0 interface is split into three 802.1Q sub-interfaces (Gi0/0.10, .20, .30), each serving as the default gateway for one VLAN. This gives inter-VLAN routing over one cable.
- **Spanning Tree Protocol (STP):** The two redundant SW1-SW2 links protect against a single link failure but would otherwise cause a switching loop and broadcast storm. STP elects a root bridge and blocks one redundant port to keep the topology loop-free. If the active link fails, the blocked port moves to forwarding and connectivity is restored.

Result: A segmented, redundant campus network where departments are isolated at Layer 2, inter-department traffic is centrally controlled by R1, and a single link failure does not cut off any department.

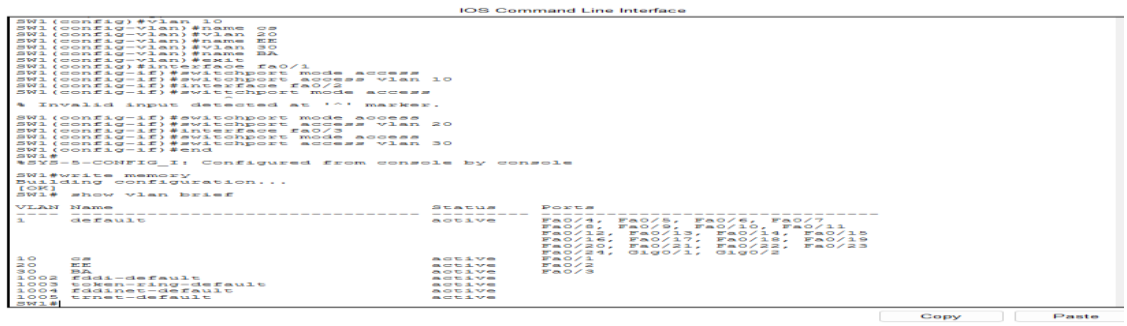
2. Final IP Addressing Table

Interface	VLAN	IP Address	Subnet Mask	Default Gateway	Connected To
Fa0	10 (CS)	192.168.10.8	255.255.255.0	192.168.10.1	SW1 Fa0/1
Fa0	10 (CS)	192.168.10.26	255.255.255.0	192.168.10.1	SW2 Fa0/1
Fa0	20 (EE)	192.168.20.8	255.255.255.0	192.168.20.1	SW1 Fa0/2
Fa0	20 (EE)	192.168.20.26	255.255.255.0	192.168.20.1	SW2 Fa0/2
Fa0	30 (BA)	192.168.30.8	255.255.255.0	192.168.30.1	SW1 Fa0/3
Fa0	30 (BA)	192.168.30.26	255.255.255.0	192.168.30.1	SW2 Fa0/3

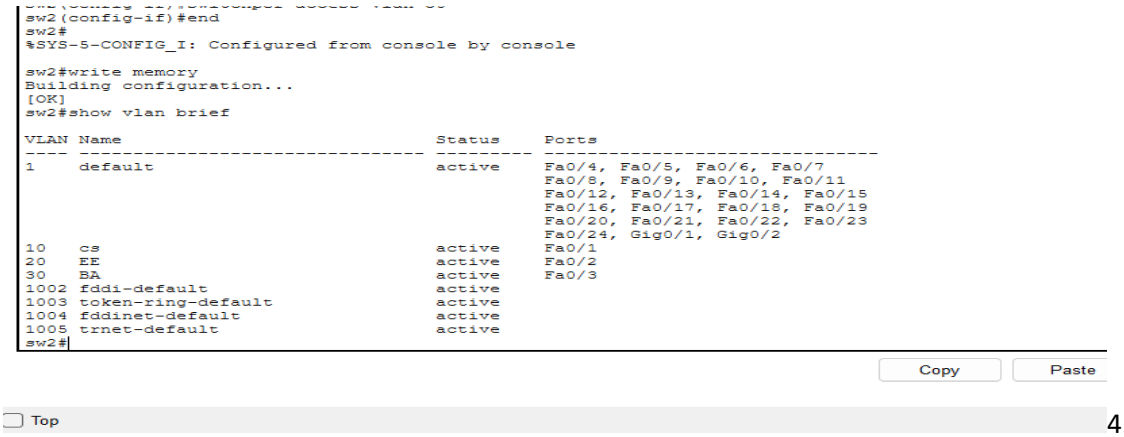
3. VLAN Configuration

VLANs 10, 20 and 30 were created on both switches and named. Fa0/1 to Fa0/3 were configured as access ports in the matching VLANs.

show vlan brief on SW1



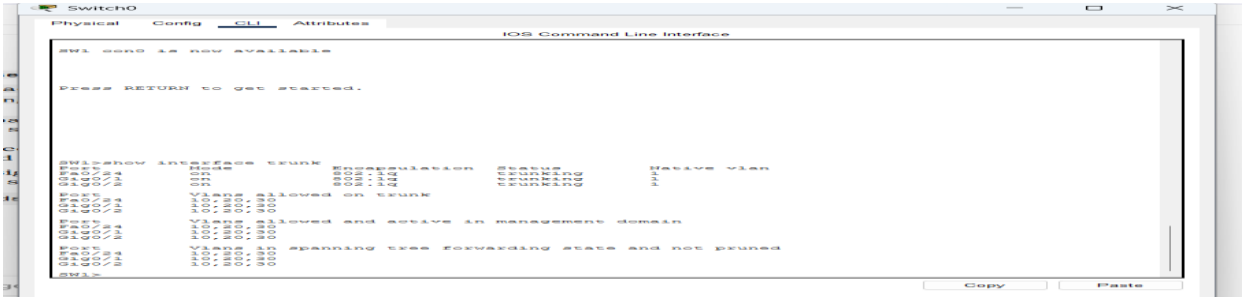
show vlan brief on SW2



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4. Trunk Status

SW1 Gi0/1, Gi0/2 and Fa0/24, and SW2 Gi0/1 and Gi0/2, operate as 802.1Q trunks in the trunking state, with VLANs 10, 20 and 30 allowed **show interfaces trunk on SW**



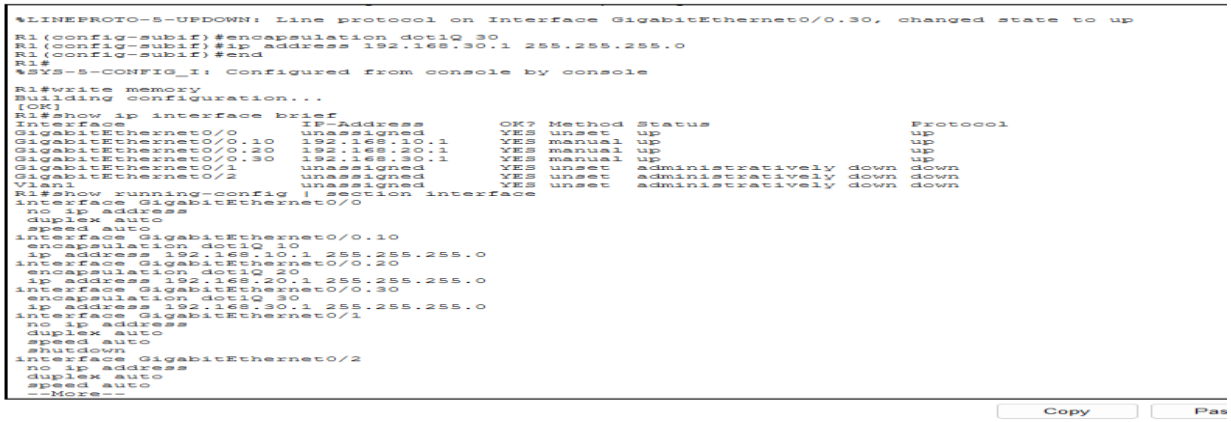
show interfaces trunk on SW2



5. Router Sub-interface Configuration

R1 Gi0/0 is enabled with no IP address. The three sub-interfaces use dot1Q encapsulation with the VLAN ID matching each sub-interface and hold the gateway addresses from the addressing table. All four interfaces show up/up.

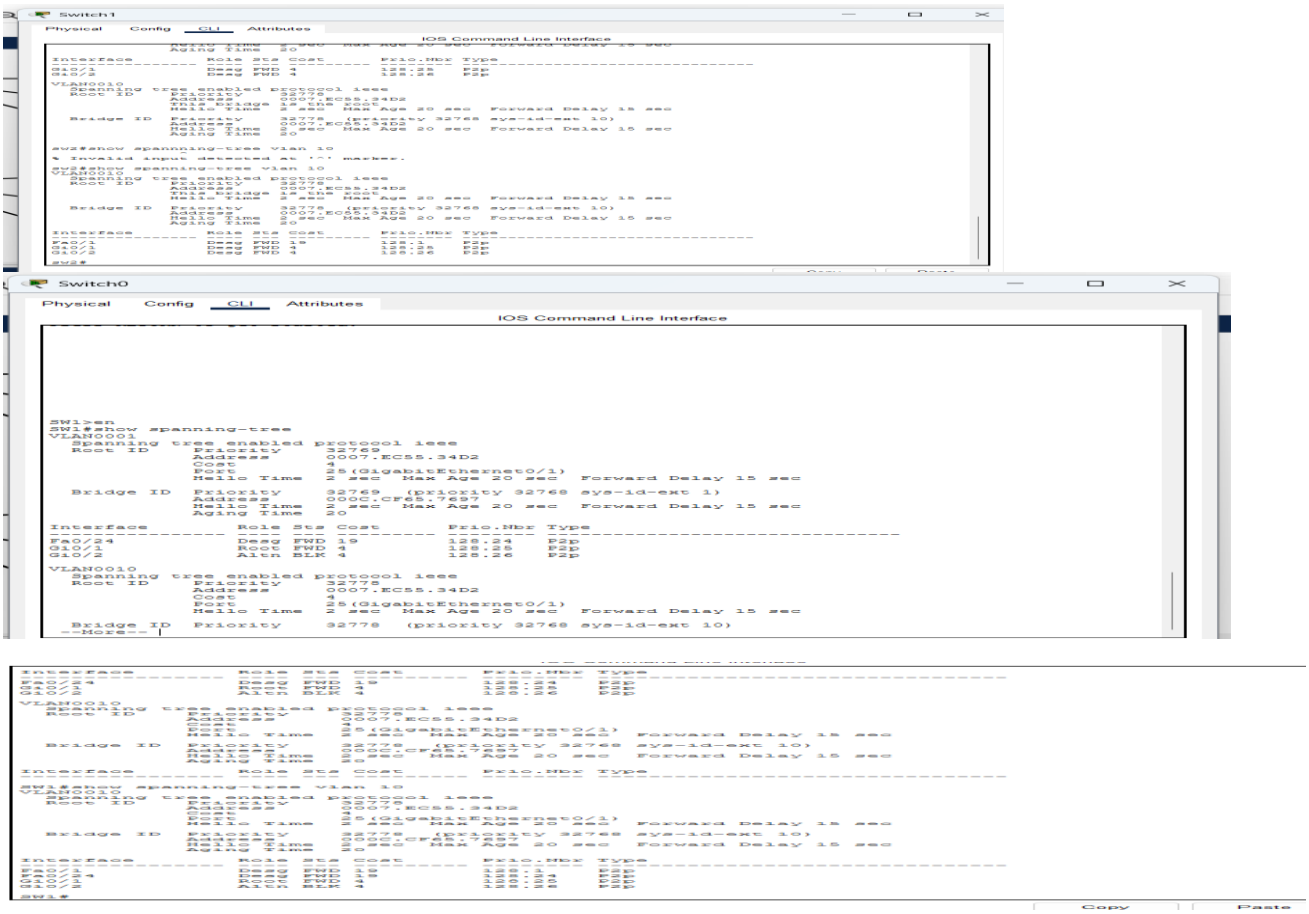
R1 sub-interface configuration (show running-config | section interface) and show ip interface brief



6. Spanning Tree Status

Default state. With default priorities (32768 on both switches), the root bridge election was decided by MAC address. SW2 (0007.EC55.34D2) had the lower MAC address and became the root. On SW1, Gi0/1 was the root port (Root/FWD), Gi0/2 was in the Alternate/Blocking state, and Fa0/24 was Designated/FWD.

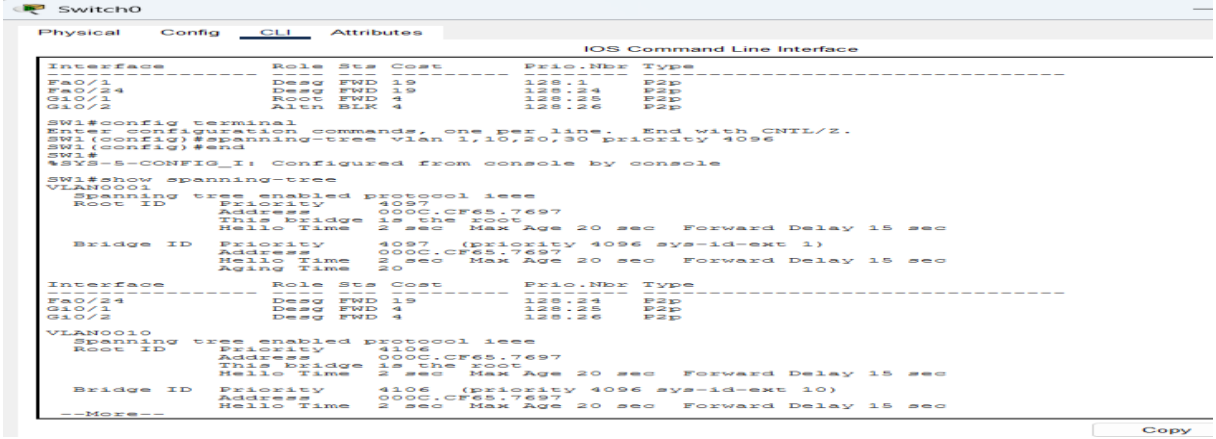
show spanning-tree on SW1 and SW2, default state (SW2 is the root, SW1 Gi0/2 is blocked)



After changing the root bridge. SW1 is connected to the router, so it was made the root bridge with the command `spanning-tree vlan 1,10,20,30 priority 4096`. SW1 (MAC 000C.CF65.7697) now reports This bridge is the root, with priority 4097 for VLAN 1 and 4106 for VLAN 10. All SW1 ports are Designated/Forwarding. On SW2, Gi0/1 is the Root port (Forwarding) and **Gi0/2 is Alternate/Blocking**, which is the port that prevents the Layer 2 loop.

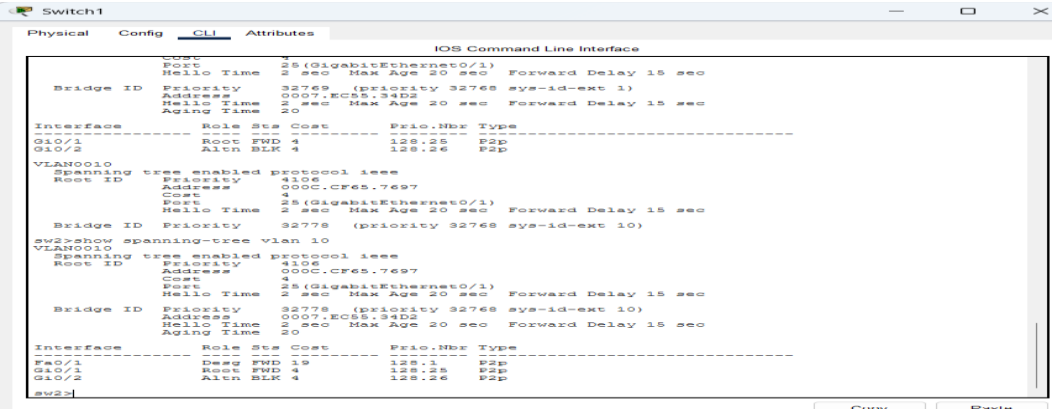
Switch	Port	Role	State
SW1 (root)	Fa0/24	Designated	Forwarding
SW1 (root)	Gi0/1	Designated	Forwarding
SW1 (root)	Gi0/2	Designated	Forwarding
SW2	Gi0/1	Root	Forwarding
SW2	Gi0/2	Alternate	**Blocking**

show spanning-tree on SW1 and SW2, after making SW1 the root





show spanning-tree vlan 10 on SW2 (Gi0/1 Root/FWD, Gi0/2 Alternate/Blocking)

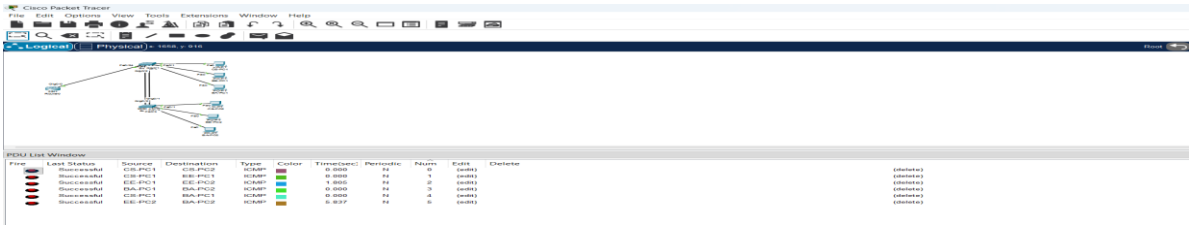


7. Verification and Testing

Ping tests were sent with the Add Simple PDU tool in Packet Tracer. All six tests were successful.

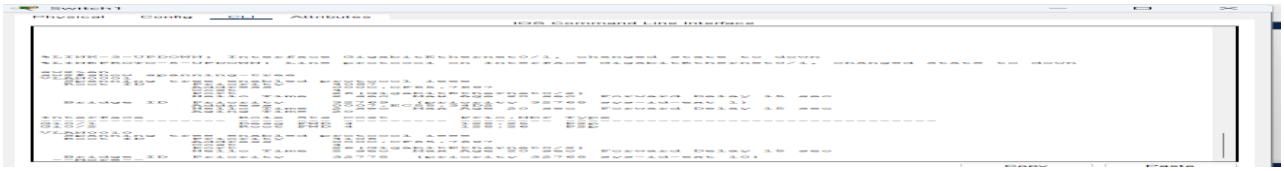
No.	Source	Destination	Type	Result
0	CS-PC1	CS-PC2	Same VLAN (10), across switches	Successful
1	CS-PC1	EE-PC1	Different VLANs, via R1	Successful
2	EE-PC1	EE-PC2	Same VLAN (20), across switches	Successful
3	BA-PC1	BA-PC2	Same VLAN (30), across switches	Successful
4	CS-PC1	BA-PC1	Different VLANs, via R1	Successful
5	EE-PC2	BA-PC2	Different VLANs, via R1	Successful

PDU List Window showing all six successful ping tests



Link failure test. SW1 Gi0/1 was shut down. SW2 moved Gi0/2 from Alternate/Blocking to Root/Forwarding, with the root ID still pointing to SW1 through Gi0/2. A new CS-PC1 to CS-PC2 ping (Scenario 1) was successful, so connectivity is maintained through the remaining link. After `no shutdown` on SW1 Gi0/1, the port went through the learning state and returned to Root/Forwarding, and Gi0/2 returned to Alternate/Blocking.

SW2 show spanning-tree during the failure, with Gi0/2 as Root/FWD



Successful ping during the failure (Scenario 1)

